

Lasso Regression

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Where this fits: Pipeline map step 8 (Model). See the [Course map](#).



- **Builds on:** Regularisation ([Note 63](#)).
- **Leads to:** Elastic Net ([Note 69](#)).
- **Compare with:** Ridge regression ([Note 66](#)); L1 and L2 regularisation in neural networks ([Note 1026](#)).

1. Overview

Key point: Lasso is Ridge with the squares in the penalty replaced by absolute values. That one change lets it push coefficients to exactly 0, so it also removes useless inputs: automatic feature selection.

Lasso regression is the second regularised version of linear regression. It is also called **L1 regularisation**. Like Ridge, it adds a penalty to the usual squared error, but the penalty uses the absolute size of each coefficient:

$$L = \sum_{i=1}^n (y_i - \hat{y}_i)^2 + \lambda \sum_{j=1}^m |\beta_j|$$

	Ridge	Lasso
Penalty	$\lambda \sum \beta_j^2$	$\lambda \sum \beta_j $
Also called	L2 regularisation	L1 regularisation
Coefficients for large λ	close to 0, never exactly 0	exactly 0

LASSO stands for “least absolute shrinkage and selection operator” (Tibshirani 1996). As with Ridge, $\lambda \geq 0$: with $\lambda = 0$ Lasso is plain linear regression, a small λ may overfit, and a large one underfits. The intercept is not penalised.

This Note shows what Lasso does with one input, with a flexible polynomial, and with the 10-input diabetes data, and why its coefficients can reach exactly 0.

2. One input: the slope reaches exactly 0

Key point: As alpha grows, the Lasso slope falls in a straight line and stops at exactly 0.

Figure 1 fits Lasso to the 100-point example of the Ridge Notes. scikit-learn calls the penalty strength **alpha**.

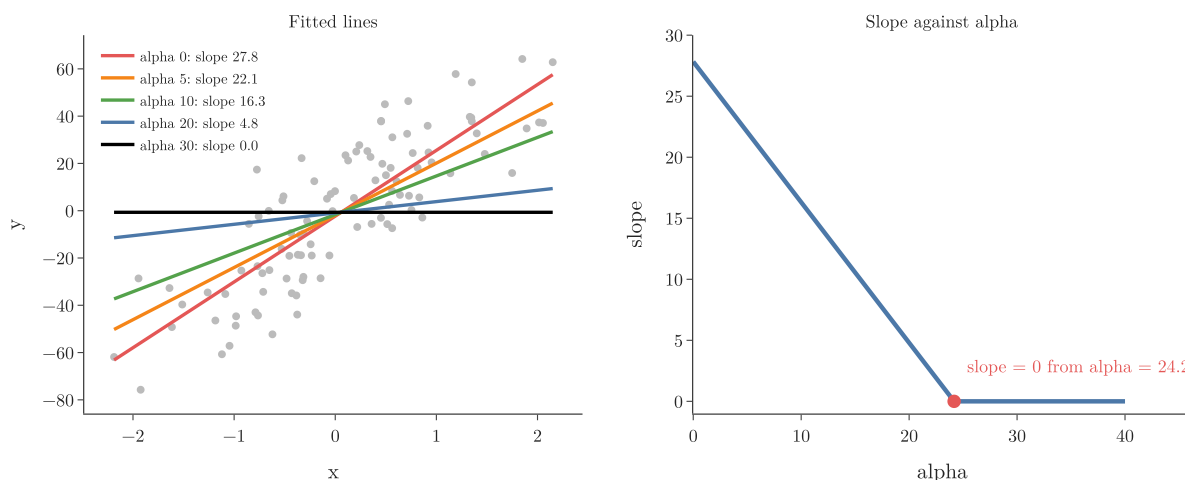


Figure 1: Lasso with one input: fitted lines and the slope against alpha

alpha	Slope	Intercept
0 (linear regression)	27.83	-2.29
5	22.07	-1.96
10	16.31	-1.62
20	4.80	-0.95
25 and above	0	-0.67

From alpha 24.17 on, the slope is exactly 0. The line is then flat at the average of y (-0.67): the input no longer plays any part. With Ridge, the slope only approached 0 (the Ridge maths Note).

Extra: scikit-learn's **Lasso** minimises $\frac{1}{2n} \sum (y_i - \hat{y}_i)^2 + \alpha \sum |\beta_j|$ (scikit-learn docs, **Lasso**): it averages the squared error and halves it. So its **alpha** is on a different scale from the λ of the formula above, and from **Ridge's alpha**. The values of alpha for Ridge and Lasso cannot be compared directly.

3. A flexible model: Lasso picks the right terms

Key point: On a degree-16 polynomial, a moderate alpha set 14 of the 16 coefficients to 0 and kept exactly x and x^2 , the true shape of the data.

Figure 2 fits a degree-16 polynomial to 100 points made from $y = 0.7x^2 - 2x + 3$ plus noise. The 16 power columns (x to x^{16}) are standardised first.

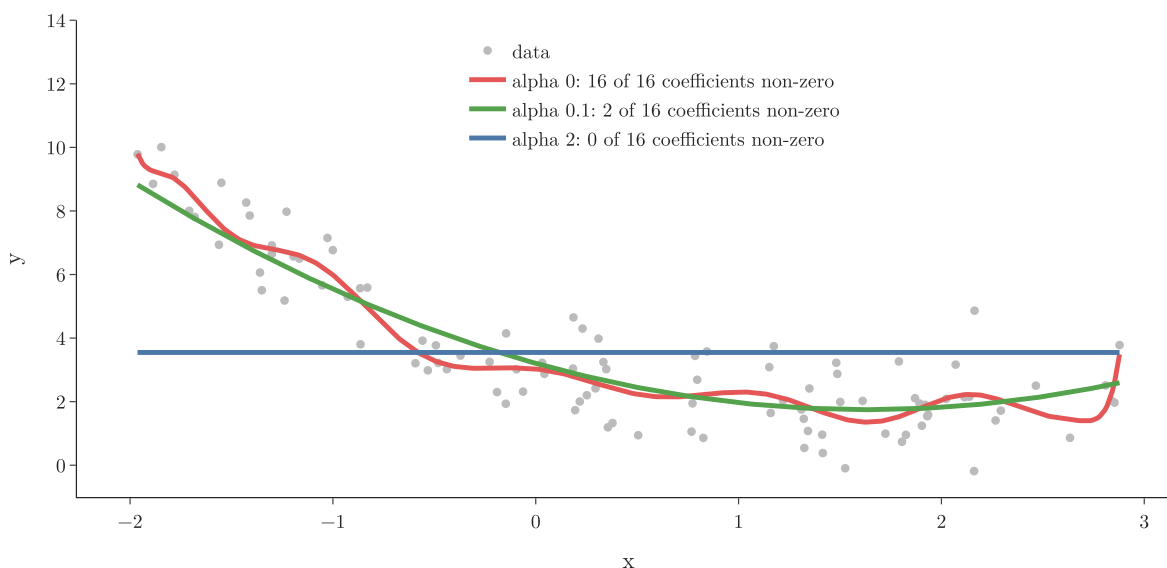


Figure 2: Degree-16 polynomial with Lasso for three values of alpha

- **alpha 0** (red): all 16 coefficients are used, and the curve wiggles to follow the noise, swinging up at the right edge: overfitting.
- **alpha 0.1** (green): only 2 coefficients are non-zero, those of x and x^2 . The curve is a smooth parabola, the true shape.
- **alpha 2** (blue): every coefficient is 0, and the model predicts the average everywhere: underfitting.

alpha	Non-zero coefficients
0.01	x, x^2, x^3, x^{16}
0.1	x, x^2
1	x

Lasso found which of the 16 columns matter without being told.

4. Feature selection

Key point: Coefficients that reach 0 remove their inputs from the model. So Lasso selects features while it trains.

A coefficient of exactly 0 means that input has no effect on the predictions, so the column can be dropped. This makes Lasso a tool for feature selection (see the [feature engineering Note](#)), useful when there are many columns and some of them do not matter.

Figure 3 trains Lasso on the 10-input diabetes data (test size 0.2, random state 2).

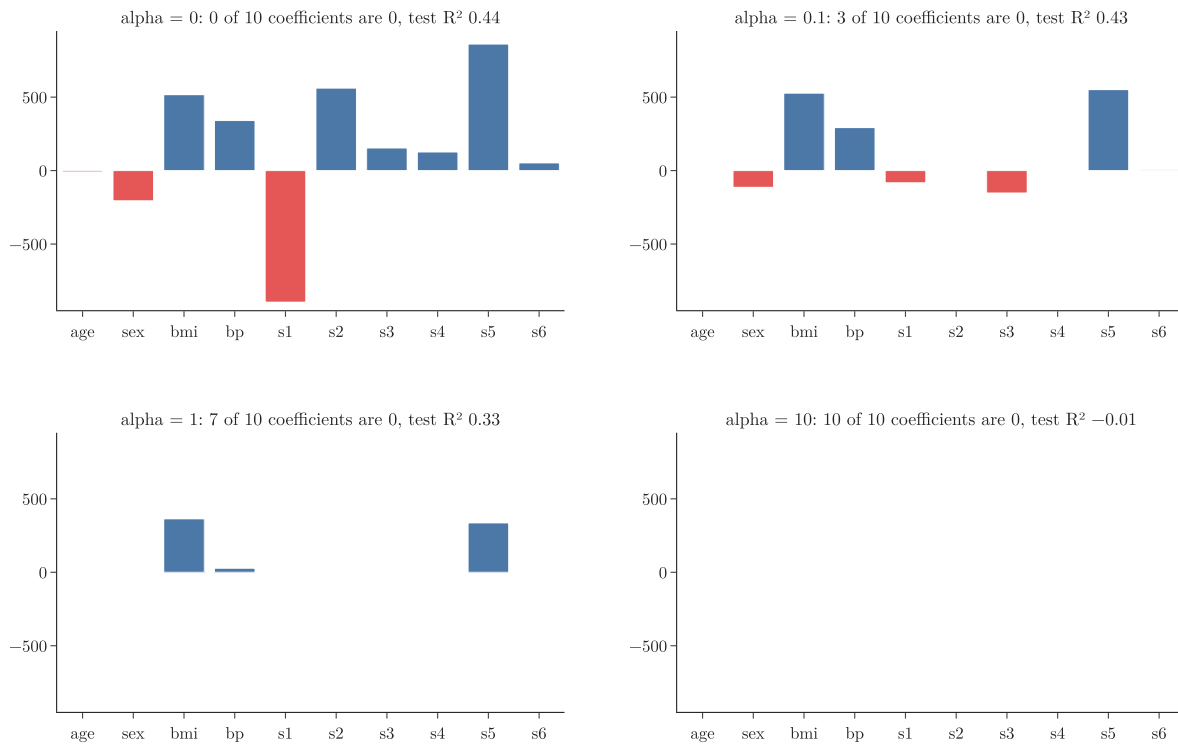


Figure 3: Diabetes coefficients for alpha 0, 0.1, 1 and 10, on the same scale

alpha	Coefficients at 0	Inputs kept	Test R^2
0	0	all 10	0.44
0.1	3	7 (age, s2, s4 dropped)	0.43
1	7	bmi, bp, s5	0.33
10	10	none	-0.01

At alpha 0.1, Lasso drops three inputs and loses almost no accuracy (0.43 against 0.44). At alpha 1, it keeps only bmi, bp and s5, the three most useful inputs here: among all 120 possible sets of three inputs, these three give plain linear regression the best cross-validated R^2 (0.485). At alpha 10, it removes everything: underfitting.

Ridge cannot do this. It shrinks an unhelpful coefficient to, say, 0.2, but the column stays in the model.

Extra: Feature selection is why Lasso is often preferred when the data has many columns, some of which are probably irrelevant: Lasso tends to do best when only a few inputs really matter (ISL §6.2.2). Fewer columns also means a simpler model that is easier to explain (ISL §6.2.2).

5. How the coefficients change

Key point: The coefficients hit 0 one at a time. The size of a coefficient does not decide when it goes: the large s2 goes second, while bmi and s5 survive longest.

Figure 4 follows each coefficient as alpha grows on a log scale.

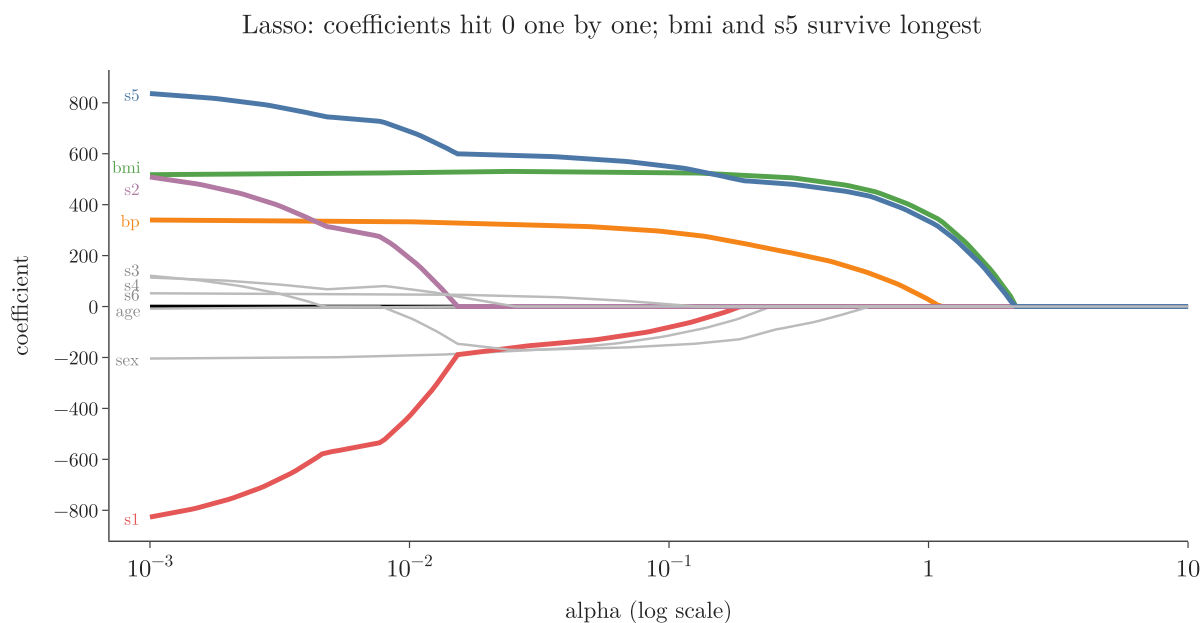


Figure 4: Lasso coefficient paths on the diabetes data

Input	Coefficient at alpha 0	Exactly 0 from alpha about
age	-9	0.012
s2	561	0.015
s4	127	0.026
s1	-896	0.20
bp	341	1.1
bmi, s5	517, 861	2.2

- As with Ridge, the very large coefficients (s1, s2, s5) are cut back hard at first. The cut barely changes the test R^2 (0.440 at alpha 0, 0.441 at alpha 0.01, 0.433 at alpha 0.1).
- Then the coefficients drop to 0 one after another. bmi and s5 are the last to go.
- s1 and s2 had huge coefficients (-896 and +561) because they are strongly correlated with each other ($r = 0.90$): such a pair can get large opposite coefficients that partly cancel out (ESL §3.4.1). Lasso drops them early anyway: size alone does not protect a coefficient.

6. Bias and variance

Key point: As with Ridge, a larger alpha raises bias and lowers variance. The best alpha is in between.

Using the same bootstrap method as the Ridge key points Note on the degree-16 polynomial (80 training points, 20 test points, 200 resamples):

alpha	Bias ² (+ noise)	Variance	Expected test error
0.001	1.45	37.34	38.79
0.01	0.74	1.24	1.98
0.1	0.80	0.06	0.86
1	2.77	0.07	2.84
3	5.38	0.07	5.45

Small alpha: huge variance (overfitting). Large alpha: high bias (underfitting). Here alpha 0.1, the value that kept only x and x^2 , has the lowest total error.

7. Why coefficients reach exactly 0

Key point: The penalty $\lambda|m|$ makes a sharp corner in the loss at $m = 0$. Once λ is large enough, that corner is the lowest point, and it stays there however large λ gets.

Take the one-input example again and hold the intercept at -2.29 , so the loss depends only on the slope:

$$L(m) = \sum_{i=1}^n (y_i - mx_i + 2.29)^2 + \lambda|m|$$

Figure 5 draws this curve while λ grows from 0 to 8,000.

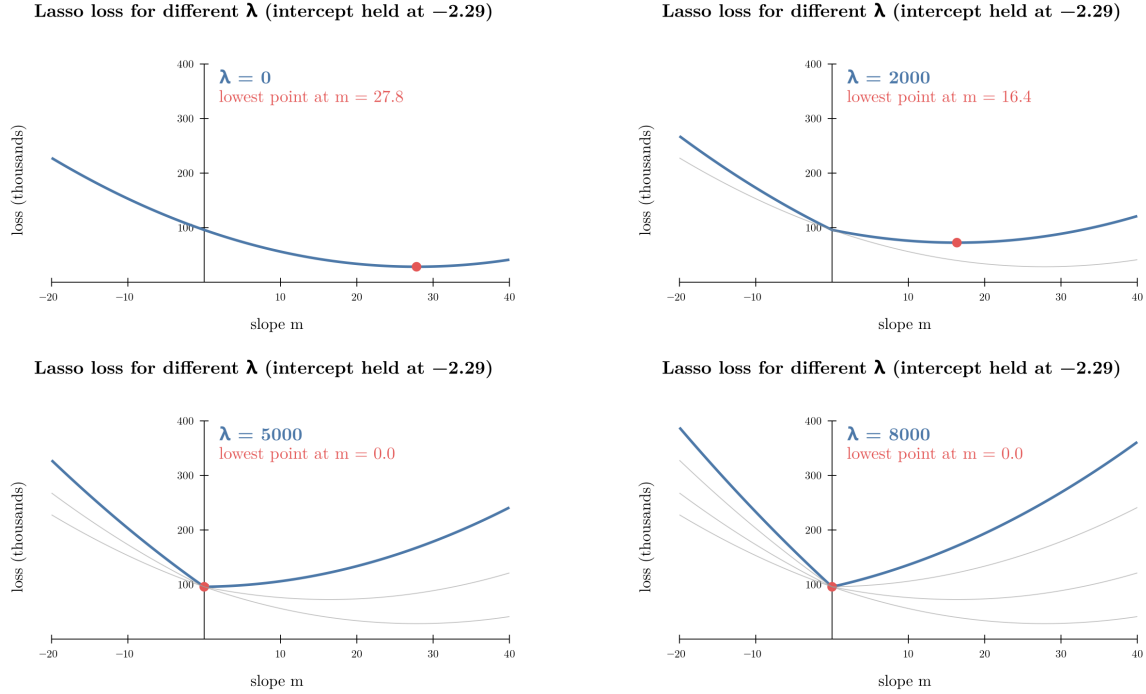


Figure 5: The Lasso loss against the slope for $\lambda = 0, 2000, 5000$ and 8000

λ	Slope at the lowest point
0	27.8
2,000	16.4
5,000	0
8,000	0

- The penalty $\lambda|m|$ is a V shape: it is 0 at $m = 0$ and grows at the same rate on both sides. Added to the smooth bowl, it gives the curve a sharp corner at $m = 0$.
- As λ grows, the lowest point moves left towards 0, as with Ridge.
- Once λ passes about 4,850 here, the corner itself is the lowest point. From then on the answer is exactly $m = 0$, and a larger λ only makes the corner steeper.

The Ridge penalty λm^2 is smooth and flat at $m = 0$, with no corner, so its lowest point only approaches 0 (the Ridge key points Note). The next Note derives this precisely.

Extra: Why do the two penalties behave so differently near 0? Look at how hard each penalty pushes, which is its slope. The slope of λm^2 is $2\lambda m$: at $m = 0.1$ it is only 0.2λ , and it fades to 0 as m approaches 0, so Ridge stops pushing. The slope of $\lambda|m|$ is λ for every positive m : it pushes with the same force all the way to 0.

8. Ridge or Lasso?

Key point: Many inputs, some probably useless: Lasso. Most inputs useful, especially correlated ones: Ridge.

	Ridge	Lasso
Penalty	squares, $\lambda \sum \beta_j^2$	absolute values, $\lambda \sum \beta_j $
Coefficients	shrink, never exactly 0	shrink, many become exactly 0
Feature selection	no	yes
Best when	most inputs matter	only some inputs matter
Closed-form formula	yes	no (solved step by step)

The “best when” row follows ISL §6.2.2. Lasso has no closed-form formula because the absolute values make the answer non-linear in y (ESL §3.4.2). With strongly correlated inputs, Lasso tends to keep only one of the group, and Ridge has been seen to predict better (Zou and Hastie 2005, §1).

Both shrink the largest coefficients, raise bias, lower variance and are tuned through λ . The difference that matters in practice is that Lasso can remove inputs. Elastic Net (a later Note) combines both penalties.

Python: Lasso in scikit-learn.

```
from sklearn.linear_model import Lasso
```

```
lasso = Lasso(alpha=0.1)
lasso.fit(X_train, y_train)
lasso.coef_           # several exactly 0
lasso.score(X_test, y_test) #  $R^2$  0.43 on the diabetes split
```

As with Ridge, inputs should be on the same scale first, for example with `make_pipeline(StandardScaler(), Lasso(alpha=0.1))`.

9. Summary

- Lasso adds $\lambda \sum |\beta_j|$ to the squared error: L1 regularisation.
- As λ grows, coefficients shrink and then become exactly 0, one by one.
- Zero coefficients remove inputs: Lasso performs feature selection (diabetes: alpha 1 keeps bmi, bp and s5; the polynomial: alpha 0.1 keeps x and x^2).
- A larger λ means more bias and less variance; too large removes every input.
- The absolute value makes a corner at 0 in the loss, which is why the answer can sit exactly at 0.

10. Sources

- **Tibshirani 1996:** Tibshirani, R. “Regression Shrinkage and Selection via the Lasso.” *Journal of the Royal Statistical Society B* 58(1), 267–288, 1996. Section 1.
- **ISL:** James, G., Witten, D., Hastie, T. and Tibshirani, R. *An Introduction to Statistical Learning*, 2nd ed. Springer, 2021. Section 6.2.2, pp. 242, 246.
- **ESL:** Hastie, T., Tibshirani, R. and Friedman, J. *The Elements of Statistical Learning*, 2nd ed. Springer, 2009. Sections 3.4.1 (p. 63) and 3.4.2 (p. 68).
- **Zou and Hastie 2005:** Zou, H. and Hastie, T. “Regularization and Variable Selection via the Elastic Net.” *Journal of the Royal Statistical Society B* 67(2), 301–320, 2005. Section 1.
- **scikit-learn docs:** `sklearn.linear_model.Lasso`, scikit-learn 1.9.

11. Key terms

Term	Meaning
Lasso regression	Linear regression with a penalty on the sum of absolute coefficients
L1 regularisation	Another name for the absolute-value penalty used by Lasso
Sparse model	A model in which many coefficients are exactly 0